FALCUTÉ DES SCIENCES ET TECHNOLOGIES (FST)

TROISIEME ANNEE

Rapport du travail de Laboratoire No 3

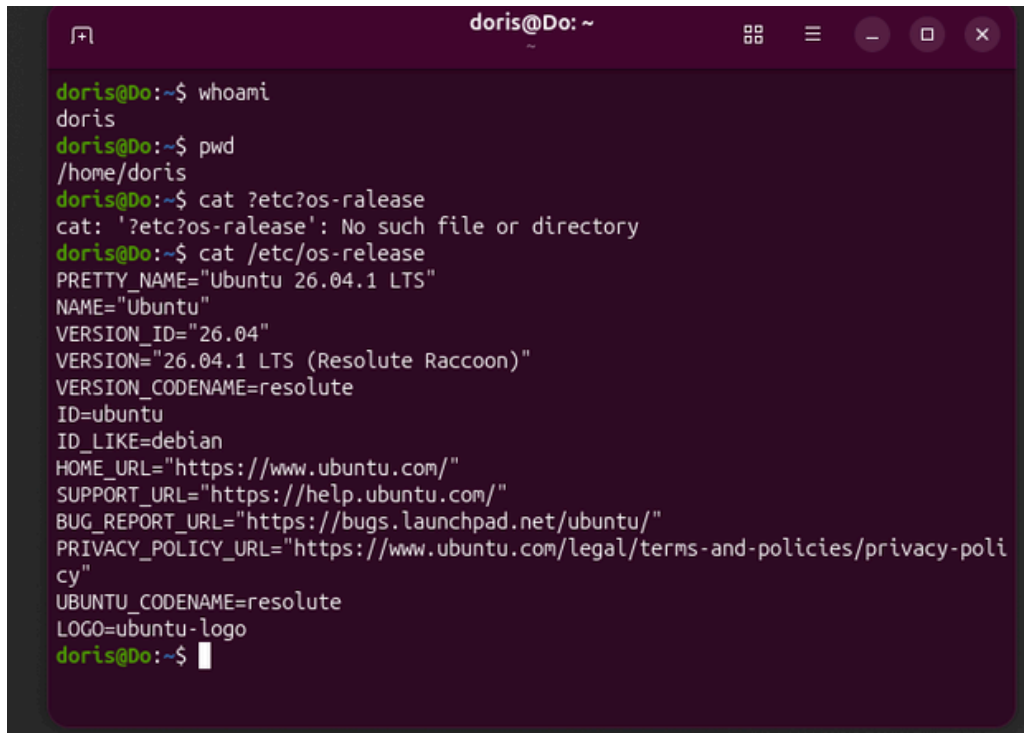
Étudiante **: Doris YOUYOUTE**

Professeur**: Ismaël SAINT AMOUR**

Le 08 Novembre 2026

Dans ce Td j’ai appris à me familiariser avec les commandes de base de Linux, la manipulation des fichiers et des répertoires, la recherche et le traitement de texte, l’observation des ressources et des processus,ainsi que la gestion des logiciels avec APT. Le TD permet également de découvrir et de comparer les environnements de bureau KDE Plasma et XFCE.

Exercice 1



La différence c’est que Un chemin absolu commence à partir de la racine / et indique l'emplacement complet d'un fichier ou dossier tandis que , un chemin relatif indique l'emplacement par rapport au répertoire courant.



```
drwxrwxr-x  2 doris doris 4096 Nov  6 12:33 resultats
drwxrwxr-x  2 doris doris 4096 Nov  6 12:33 sauvegardes
doris@Do:~$ cd ~/td_linux/documents
doris@Do:~/td_linux/documents$ printf '%s\n' \ "Travaux sur linux" \ "mes comman
des" \ "ubuntu" >notes.txt
doris@Do:~/td_linux/documents$ ls
notes.txt
doris@Do:~/td_linux/documents$ cd ..
doris@Do:~/td_linux$ cp documents/notes.txt sauvegardes/notes_initiales.txt
doris@Do:~/td_linux$ diff documents/notes.txt sauvegardes/notes_initiales.txt
doris@Do:~/td_linux$ mkdir temporaire
doris@Do:~/td_linux$ rmdir temporaire
doris@Do:~/td_linux$ touch brouillon
doris@Do:~/td_linux$ rm -i brouillon
rm: remove regular empty file 'brouillon'? y
doris@Do:~/td_linux$ echo -e "web;actif\nssh;actif\nimpressiom;inactif\nsauvegar
de;actif\nweb;actif . ~/td_linux/documents/services.txt
> ^C
doris@Do:~/td_linux$ echo -e "web;actif\nssh;actif\nimpressiom;inactif\nsauvegar
de;actif\nweb;actif > ~/td_linux/documents/services.txt
> ^C
doris@Do:~/td_linux$ echo -e "web;actif\nssh;actif\nimpressiom;inactif\nsauvegar
de;actif\nweb;actif" > ~/td_linux/documents/services.txt
```

```
doris@Do:~/td_linux$ echo -e "web;actif\nssh;actif\nimpressiom;inactif\nsauvegar
de;actif\nweb;actif" > ~/td_linux/documents/services.txt
doris@Do:~/td_linux$ cat ~/td_linux/documents/services.txt
web;actif
ssh;actif
impressiom;inactif
sauvegarde;actif
web;actif
doris@Do:~/td_linux$ cd documents
doris@Do:~/td_linux/documents$ head -2 services.txt
web;actif
ssh;actif
doris@Do:~/td_linux/documents$ tail -2 services.txt
sauvegarde;actif
web;actif
doris@Do:~/td_linux/documents$ wc -l services.txt
5 services.txt
doris@Do:~/td_linux/documents$
```

```
doris@Do:~/td_linux/documents$ tar -czvf ../sauvegardes/documents.tar.gz .
./
./notes.txt
./services.txt
doris@Do:~/td_linux/documents$ tar -tzvf ../sauvegardes/documents.tar.gz
drwxrwxr-x doris/doris       0 2026-11-06 13:08 ./
-rw-rw-r-- doris/doris      42 2026-11-06 12:40 ./notes.txt
-rw-rw-r-- doris/doris      66 2026-11-06 13:08 ./services.txt
doris@Do:~/td_linux/documents$ mkdir restauration
doris@Do:~/td_linux/documents$ tar -xzvf ../sauvegardes/documents.tar.gz -C rest
auration/
./
./notes.txt
./services.txt
doris@Do:~/td_linux/documents$ diff services.txt restauration/services.txt
doris@Do:~/td_linux/documents$
```

```
doris@Do:~/td_linux/documents$ diff services.txt restauration/services.txt
doris@Do:~/td_linux/documents$ date
Fri Nov  6 01:43:30 PM EST 2026
doris@Do:~/td_linux/documents$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           875M  2.4M  873M   1% /run
/dev/sda2        24G   11G   12G  50% /
tmpfs           2.2G     0  2.2G   0% /dev/shm
efivarfs        256K   29K  223K  12% /sys/firmware/efi/efivars
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
none            1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
tmpfs           2.2G  8.0K  2.2G   1% /tmp
/dev/sda1       1.1G  6.4M  1.1G   1% /boot/efi
tmpfs           438M   84K  438M   1% /run/user/1000
doris@Do:~/td_linux/documents$ du -sh ~/td_linux
44K     /home/doris/td_linux
doris@Do:~/td_linux/documents$ free -h
               total        used        free      shared  buff/cache   available
Mem:           4.3Gi       1.2Gi       1.5Gi        24Mi       1.8Gi       3.0Gi
Swap:          4.0Gi          0B       4.0Gi
doris@Do:~/td_linux/documents$
```

```
Swap:            4.0Gi          0B        4.0Gi
doris@Do:~/td_linux/documents$ sleep 300 &
[1] 6262
doris@Do:~/td_linux/documents$ PID=$!
doris@Do:~/td_linux/documents$ ps -p $PID
    PID TTY          TIME CMD
   6262 pts/0    00:00:00 sleep
doris@Do:~/td_linux/documents$ kill $PID
[1]+  Terminated                 sleep 300
doris@Do:~/td_linux/documents$ ps -p $PID
    PID TTY          TIME CMD
doris@Do:~/td_linux/documents$ top
```

```
Tasks: 198 total,   1 running, 197 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.9 us,  1.0 sy,  0.0 ni, 95.7 id,  0.0 wa,  0.0 hi,  2.4 si,  0.0 st
MiB Mem :   4372.9 total,   1560.7 free,   1269.4 used,   1795.0 buff/cache
MiB Swap:   4096.0 total,   4096.0 free,      0.0 used.   3103.5 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   2525 doris     20   0 3960560 359072 163964 S   4.0   8.0   0:52.91 gnome-s+
   3455 doris     20   0 1804204 394600 136828 S   1.7   8.8   0:29.26 ptyxis
   5369 root      20   0       0      0      0 I   0.3   0.0   0:00.79 kworker+
      1 root      20   0   26364  17376  11584 S   0.0   0.4   0:04.50 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_wo+
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
     12 root      20   0       0      0      0 I   0.0   0.0   0:00.00 kworker+
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
     14 root      20   0       0      0      0 S   0.0   0.0   0:13.95 ksoftir+
     15 root      20   0       0      0      0 I   0.0   0.0   0:00.99 rcu_pre+
     16 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp+
     17 root      20   0       0      0      0 S   0.0   0.0   0:00.04 rcu_exp+
doris@Do:~/td_linux/documents$
```

```
/etc/kernel/postinst.d/zz-update-grub:
Sourcing file `/etc/default/grub'
Sourcing file `/etc/default/grub.d/kdump-tools.cfg'
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-7.0.0-43-generic
Found initrd image: /boot/initrd.img-7.0.0-43-generic
Found linux image: /boot/vmlinuz-7.0.0-42-generic
Found initrd image: /boot/initrd.img-7.0.0-42-generic
Found linux image: /boot/vmlinuz-7.0.0-30-generic
Found initrd image: /boot/initrd.img-7.0.0-30-generic
Found memtest86+ 64bit EFI image: /boot/mt86+x64
Found memtest86+ 32bit EFI image: /boot/mt86+ia32
Found memtest86+ 64bit image: /boot/mt86+x64
Found memtest86+ 32bit image: /boot/mt86+ia32
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done
Processing triggers for dracut (110-11ubuntu0.1)…
update-initramfs: Generating /boot/initrd.img-7.0.0-43-generic
doris@Do:~/td_linux/documents$
```

```
doris@Do: ~/td_linux/documents — apt search htop
~/td_linux/documents

aha/resolute 0.5.1-3build2 amd64
  ANSI color to HTML converter

bashtop/resolute 0.9.25-1build1 all
  Resource monitor that shows usage and stats

bpytop/resolute 1.0.68-2build1 all
  Resource monitor that shows usage and stats

btm/resolute 0.12.3+~0+20250330+~0.30.0+~0.37.2+~0.3.5-6 amd64
  customizable graphical process/system monitor for the terminal

btop/resolute 1.4.6-2 amd64
  Modern and colorful command line resource monitor that shows usage and stats

htop/resolute 3.4.1-5build2 amd64
  interactive processes viewer

libauthen-oath-perl/resolute 2.0.1-2 all
  Perl module for OATH One Time Passwords

neovim-toggleterm/resolute 2.13.1-1 all
  Plugin that creates and toggles persistent terminals on neovim
lines 1-23
```

```
Package: htop
Version: 3.4.1-5build2
Priority: optional
Section: utils
Origin: Ubuntu
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Original-Maintainer: Daniel Lange <DLange@debian.org>
Bugs: https://bugs.launchpad.net/ubuntu/+filebug
Installed-Size: 451 kB
Depends: libc6 (>= 2.38), libncursesw6 (>= 6), libtinfo6 (>= 6)
Suggests: lm-sensors, lsof, strace
Homepage: https://htop.dev/
Task: cloud-image, cloud-image, server, ubuntu-server-raspi, kubuntu-desktop, k>
Download-Size: 177 kB
APT-Sources: http://ht.archive.ubuntu.com/ubuntu resolute/main amd64 Packages
Description: interactive processes viewer
 Htop is an ncursed-based process viewer similar to top, but it
 allows one to scroll the list vertically and horizontally to see
 all processes and their full command lines.
 .
 Tasks related to processes (killing, renicing) can be done without
 entering their PIDs.

doris@Do:~/td_linux/documents$
```

```
doris@Do: ~/td_linux/documents
~/td_linux/documents

Processing triggers for hicolor-icon-theme (0.18-2build1)…
Processing triggers for gnome-menus (3.38.1-1ubuntu1)…
Processing triggers for man-db (2.13.1-1build1)…
doris@Do:~/td_linux/documents$ dpkg -s htop
Package: htop
Status: install ok installed
Priority: optional
Section: utils
Installed-Size: 440
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Architecture: amd64
Version: 3.4.1-5build2
Depends: libc6 (>= 2.38), libncursesw6 (>= 6), libtinfo6 (>= 6)
Suggests: lm-sensors, lsof, strace
Description: interactive processes viewer
 Htop is an ncursed-based process viewer similar to top, but it
 allows one to scroll the list vertically and horizontally to see
 all processes and their full command lines.
 .
 Tasks related to processes (killing, renicing) can be done without
 entering their PIDs.
Original-Maintainer: Daniel Lange <DLange@debian.org>
Homepage: https://htop.dev/
doris@Do:~/td_linux/documents$
```

```
doris@Do: ~/td_linux/documents
~/td_linux/documents

doris@Do:~/td_linux/documents$ apt -s install htop
NOTE: This is only a simulation!
      apt needs root privileges for real execution.
      Keep also in mind that locking is deactivated,
      so don't depend on the relevance to the real current situation!

The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-30                  linux-modules-7.0.0-30-generic
  linux-headers-7.0.0-30-generic          linux-tools-7.0.0-30
  linux-image-unsigned-7.0.0-30-generic   linux-tools-7.0.0-30-generic
  linux-main-modules-zfs-7.0.0-30-generic
Use 'apt autoremove' to remove them.

Installing:
  htop

Suggested packages:
  lm-sensors

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 3
Inst htop (3.4.1-5build2 Ubuntu:26.04/resolute [amd64])
Conf htop (3.4.1-5build2 Ubuntu:26.04/resolute [amd64])
doris@Do:~/td_linux/documents$
```

```
~/td_linux/documents

  0[||                                    2.6%] Tasks: 112, 368 thr, 88 kthr; 1 runnin
  1[                                      0.0%] Load average: 0.21 0.12 0.10
Mem[|||||||||||||||||||1007M/4.27G] Uptime: 00:15:28
Swp[                                    0K/4.00G]

  Main   I/O
    PID USER       PRI  NI  VIRT   RES   SHR S  CPU%▽MEM%   TIME+  Command
   3898 doris       20   0 11472  5204  3940 R   2.1  0.1  0:00.15 htop
   2515 doris       20   0 3854M  329M     0 S   0.7  7.5  0:03.49 /usr/bin/gnom
   3253 doris       20   0  483M 29112 21672 S   0.7  0.7  0:00.11 /usr/libexec/
   3274 doris       20   0 1743M  358M   131M S   0.7  8.2  0:04.19 /usr/bin/ptyx
      1 root        20   0 25512 16840 11824 S   0.0  0.4  0:02.07 /usr/lib/syst
    357 root        19  -1 51076 21248 19496 S   0.0  0.5  0:00.36 /usr/lib/syst
    405 systemd-oo  20   0 16824  7744  6704 S   0.0  0.2  0:00.42 /usr/lib/syst
    406 systemd-re  20   0 22444 14992 12408 S   0.0  0.3  0:00.13 /usr/lib/syst
    410 root        20   0 41848 15328  9372 S   0.0  0.3  0:00.54 /usr/lib/syst
   1025 avahi       20   0  6752  4872  4488 S   0.0  0.1  0:00.08 avahi-daemon:
   1026 root        20   0  2888  2000  1884 S   0.0  0.0  0:00.05 /bin/sh /usr/
   1027 messagebus  20   0 11908  8724  4812 S   0.0  0.2  0:00.62 @dbus-daemon
   1034 root        20   0 49444 30220 14944 S   0.0  0.7  0:00.24 /usr/bin/pyth
   1036 polkitd     20   0  374M 11520  8232 S   0.0  0.3  0:00.39 /usr/lib/polk
   1046 root        20   0 1845M 46520 31080 S   0.0  1.0  0:00.08 /usr/lib/snap
F1Help  F2Setup F3SearchF4FilterF5Tree  F6SortByF7Nice -F8Nice +F9Kill  F10Quit
```

```
doris@Do: ~/td_linux/documents
~/td_linux/documents

The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-30                  linux-modules-7.0.0-30-generic
  linux-headers-7.0.0-30-generic          linux-tools-7.0.0-30
  linux-image-unsigned-7.0.0-30-generic   linux-tools-7.0.0-30-generic
  linux-main-modules-zfs-7.0.0-30-generic
Use 'sudo apt autoremove' to remove them.

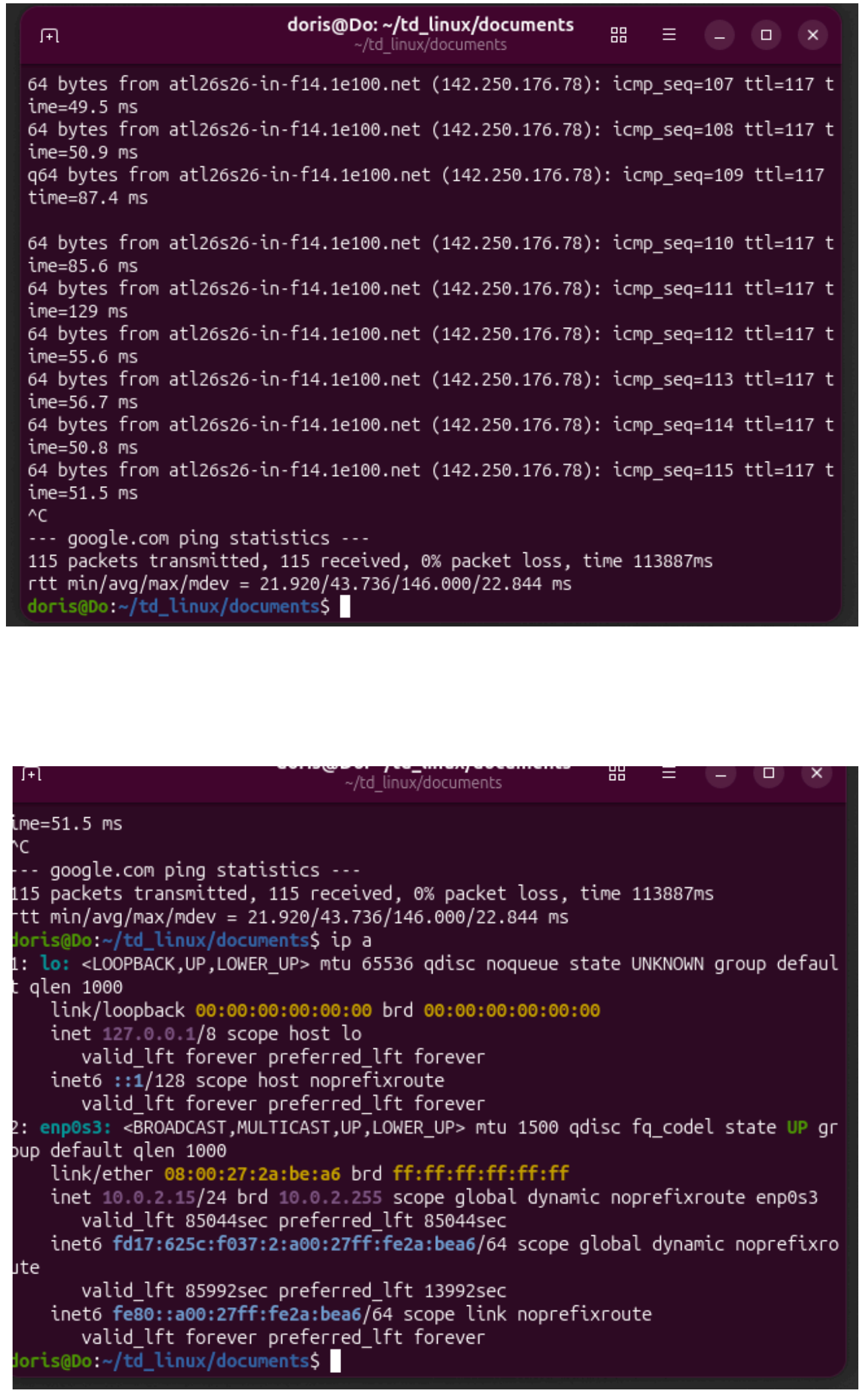
REMOVING:
  htop

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 3
  Freed space: 451 kB

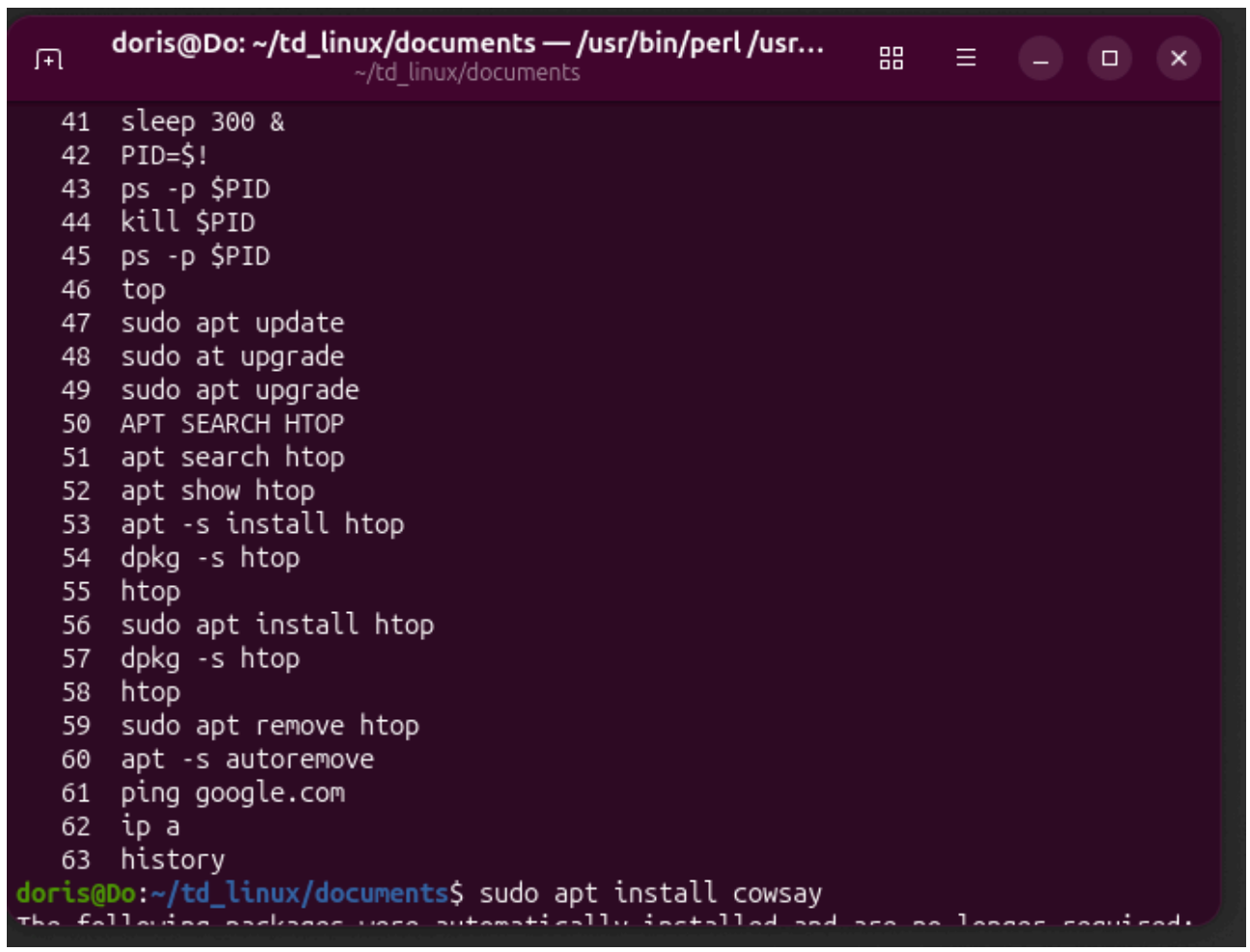
Continue? [Y/n] y
(Reading database… 236111 files and directories currently installed.)
Removing htop (3.4.1-5build2)…
Processing triggers for hicolor-icon-theme (0.18-2build1)…
Processing triggers for gnome-menus (3.38.1-1ubuntu1)…
Processing triggers for man-db (2.13.1-1build1)…
Processing triggers for desktop-file-utils (0.28-1build1)…
doris@Do:~/td_linux/documents$
```

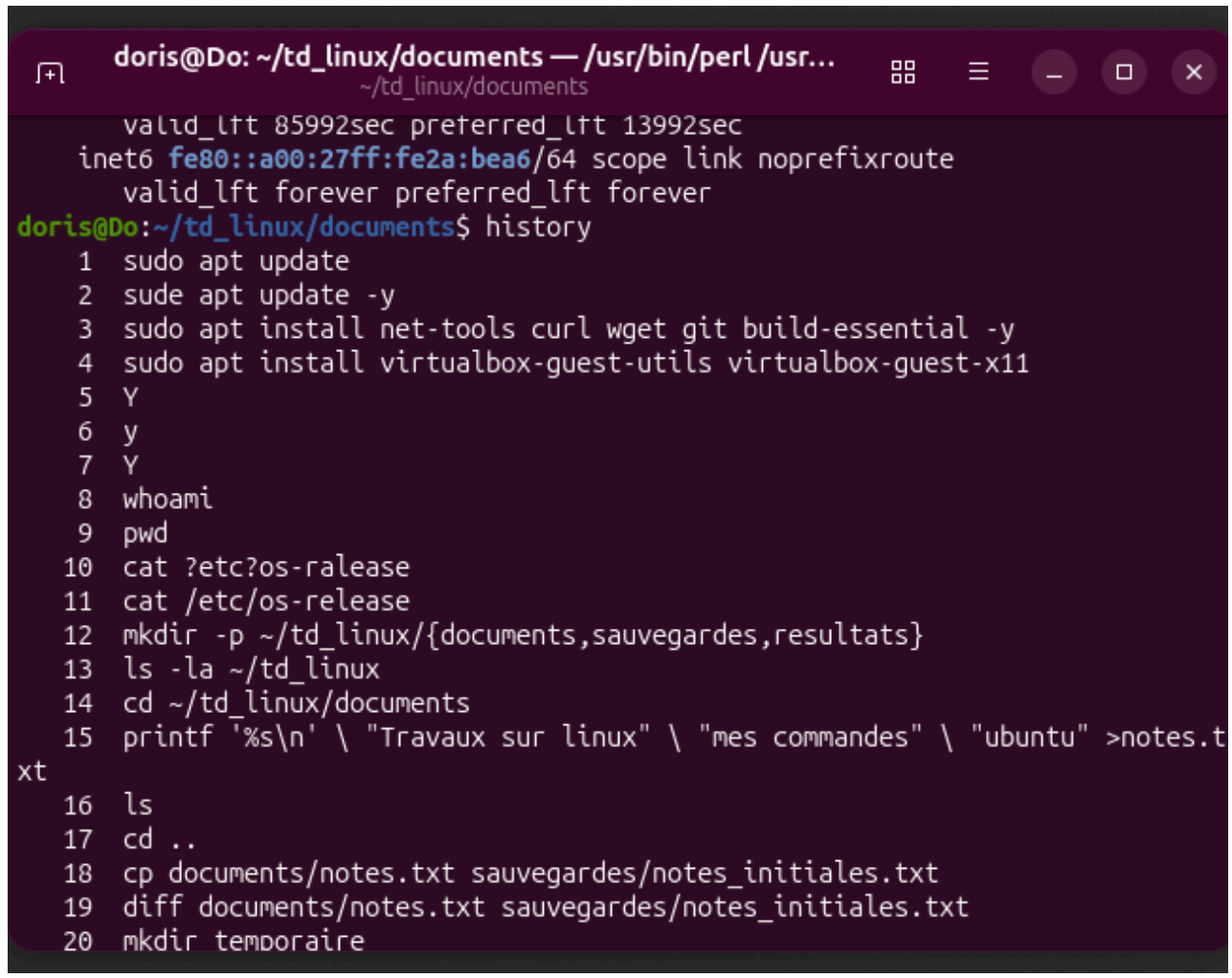
```
doris@Do: ~/td_linux/documents — ping google.com
~/td_linux/documents

Remv linux-tools-7.0.0-30 [7.0.0-30.30]
doris@Do:~/td_linux/documents$ ping google.com
PING google.com (142.250.176.78) 56(84) bytes of data.
64 bytes from atl26s26-in-f14.1e100.net (142.250.176.78): icmp_seq=1 ttl=117 tim
e=37.9 ms
64 bytes from atl26s26-in-f14.1e100.net (142.250.176.78): icmp_seq=2 ttl=117 tim
e=34.8 ms
64 bytes from atl26s26-in-f14.1e100.net (142.250.176.78): icmp_seq=3 ttl=117 tim
e=34.4 ms
64 bytes from atl26s26-in-f14.1e100.net (142.250.176.78): icmp_seq=4 ttl=117 tim
e=33.1 ms
64 bytes from atl26s26-in-f14.1e100.net (142.250.176.78): icmp_seq=5 ttl=117 tim
e=35.1 ms
64 bytes from atl26s26-in-f14.1e100.net (142.250.176.78): icmp_seq=6 ttl=117 tim
e=31.2 ms
64 bytes from atl26s26-in-f14.1e100.net (142.250.176.78): icmp_seq=7 ttl=117 tim
e=32.7 ms
64 bytes from atl26s26-in-f14.1e100.net (142.250.176.78): icmp_seq=8 ttl=117 tim
e=33.6 ms
64 bytes from atl26s26-in-f14.1e100.net (142.250.176.78): icmp_seq=9 ttl=117 tim
e=42.8 ms
64 bytes from atl26s26-in-f14.1e100.net (142.250.176.78): icmp_seq=10 ttl=117 ti
me=40.4 ms
64 bytes from atl26s26-in-f14.1e100.net (142.250.176.78): icmp_seq=11 ttl=117 ti
me=33.4 ms
```


doris@Do: ~/td_linux/documents
~/td_linux/documents
64 bytes from atl26s26-in-f14.1e100.net (142.250.176.78): icmp_seq=107 ttl=117 time=49.5 ms
64 bytes from atl26s26-in-f14.1e100.net (142.250.176.78): icmp_seq=108 ttl=117 time=50.9 ms
q64 bytes from atl26s26-in-f14.1e100.net (142.250.176.78): icmp_seq=109 ttl=117 time=87.4 ms
64 bytes from atl26s26-in-f14.1e100.net (142.250.176.78): icmp_seq=110 ttl=117 time=85.6 ms
64 bytes from atl26s26-in-f14.1e100.net (142.250.176.78): icmp_seq=111 ttl=117 time=129 ms
64 bytes from atl26s26-in-f14.1e100.net (142.250.176.78): icmp_seq=112 ttl=117 time=55.6 ms
64 bytes from atl26s26-in-f14.1e100.net (142.250.176.78): icmp_seq=113 ttl=117 time=56.7 ms
64 bytes from atl26s26-in-f14.1e100.net (142.250.176.78): icmp_seq=114 ttl=117 time=50.8 ms
64 bytes from atl26s26-in-f14.1e100.net (142.250.176.78): icmp_seq=115 ttl=117 time=51.5 ms
^C
--- google.com ping statistics ---
115 packets transmitted, 115 received, 0% packet loss, time 113887ms
rtt min/avg/max/mdev = 21.920/43.736/146.000/22.844 ms
doris@Do:~/td_linux/documents$

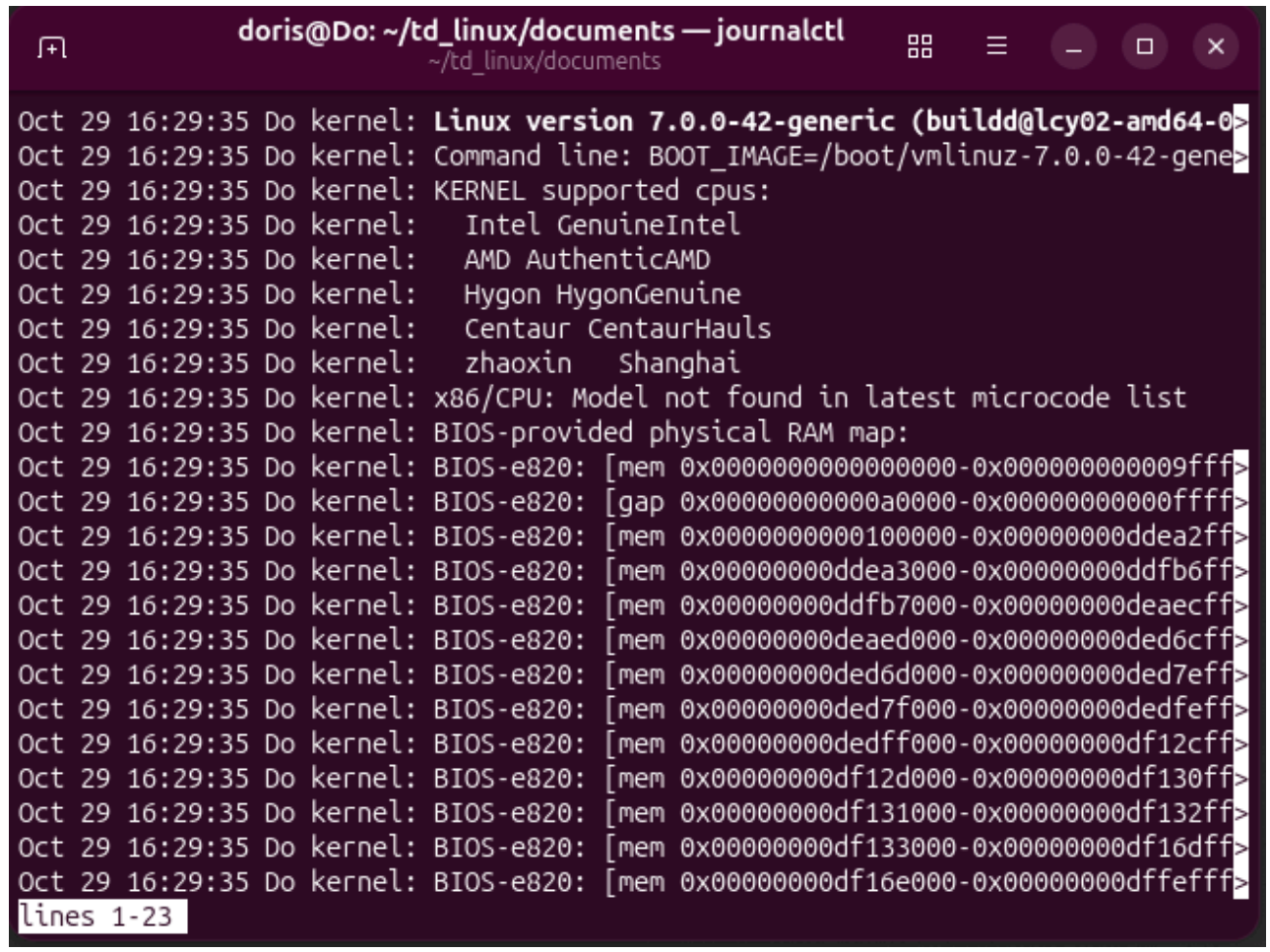
~/td_linux/documents
ime=51.5 ms
^C
--- google.com ping statistics ---
115 packets transmitted, 115 received, 0% packet loss, time 113887ms
rtt min/avg/max/mdev = 21.920/43.736/146.000/22.844 ms
doris@Do:~/td_linux/documents$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:00:27:2a:be:a6 brd ff:ff:ff:ff:ff:ff
inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
valid_lft 85044sec preferred_lft 85044sec
inet6 fd17:625c:f037:2:a00:27ff:fe2a:bea6/64 scope global dynamic noprefixroute
valid_lft 85992sec preferred_lft 13992sec
inet6 fe80::a00:27ff:fe2a:bea6/64 scope link noprefixroute
valid_lft forever preferred_lft forever
doris@Do:~/td_linux/documents$

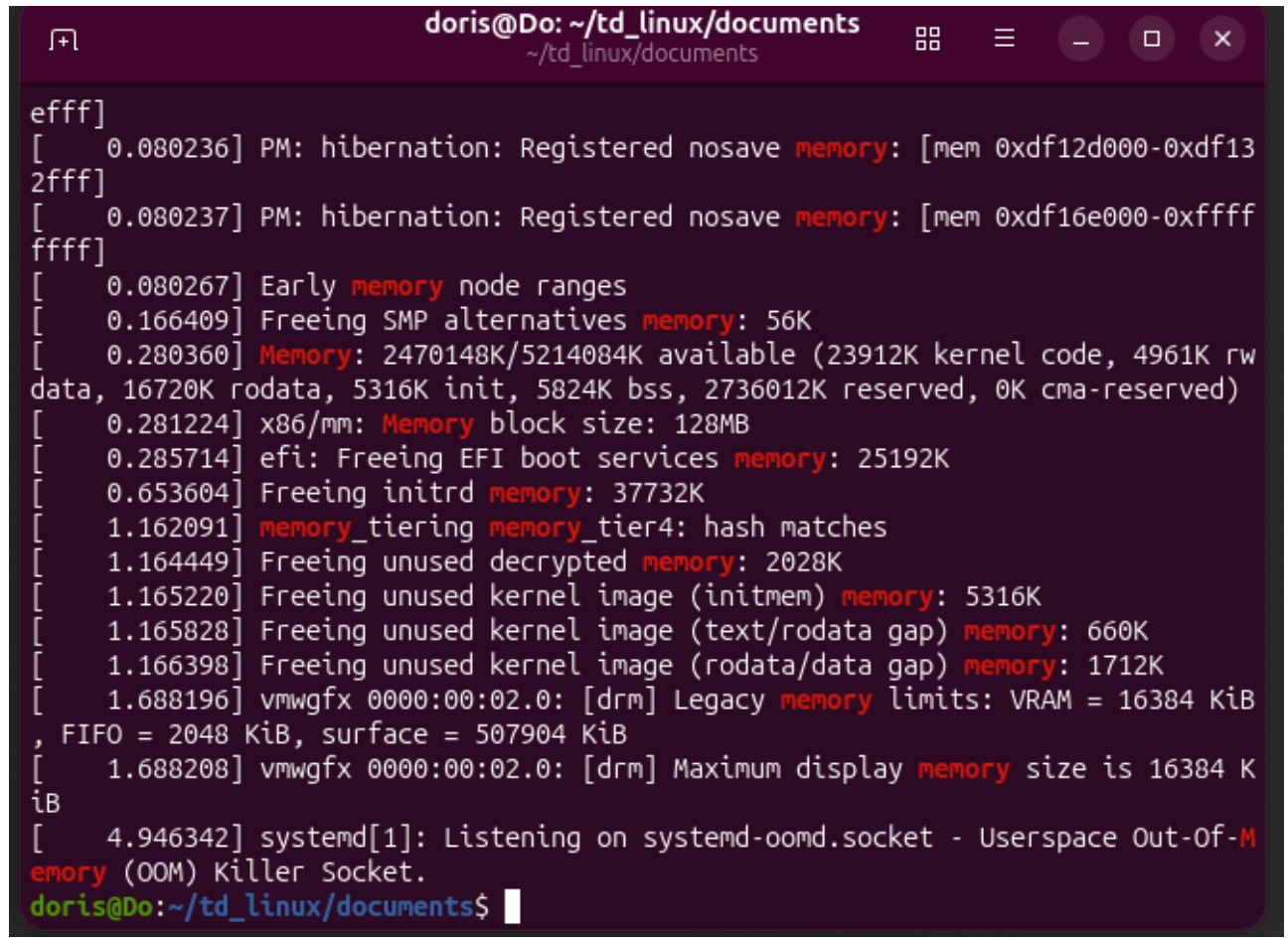

doris@Do: ~/td_linux/documents — /usr/bin/perl /usr...
~/td_linux/documents
41 sleep 300 &
42 PID=$!
43 ps -p $PID
44 kill $PID
45 ps -p $PID
46 top
47 sudo apt update
48 sudo at upgrade
49 sudo apt upgrade
50 APT SEARCH HTOP
51 apt search htop
52 apt show htop
53 apt -s install htop
54 dpkg -s htop
55 htop
56 sudo apt install htop
57 dpkg -s htop
58 htop
59 sudo apt remove htop
60 apt -s autoremove
61 ping google.com
62 ip a
63 history
doris@Do:~/td_linux/documents$ sudo apt install cowsay

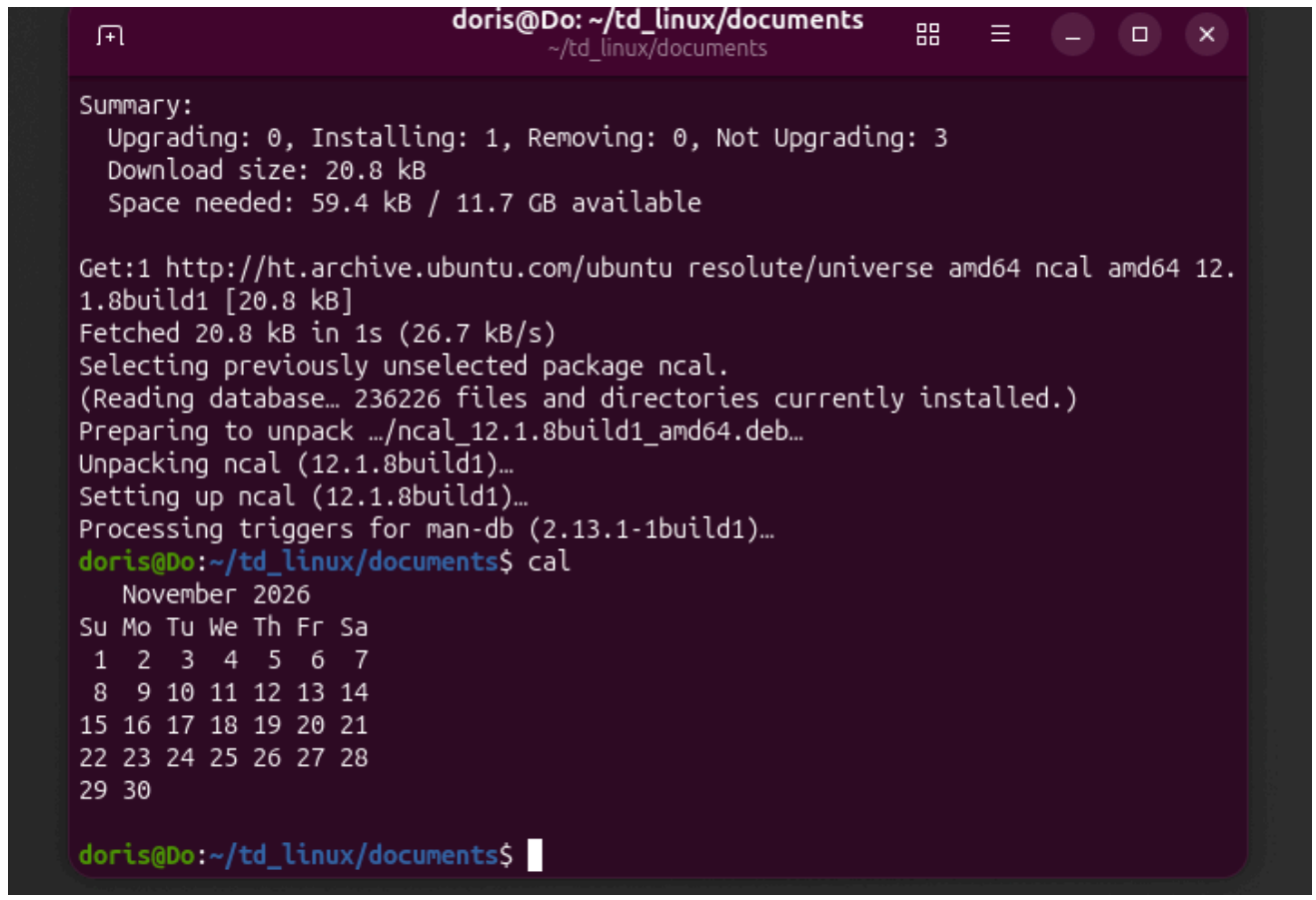

doris@Do: ~/td_linux/documents — /usr/bin/perl /usr...
~/td_linux/documents
valid_lft 85992sec preferred_lft 13992sec
inet6 fe80::a00:27ff:fe2a:bea6/64 scope link noprefixroute
valid_lft forever preferred_lft forever
doris@Do:~/td_linux/documents$ history
1 sudo apt update
2 sude apt update -y
3 sudo apt install net-tools curl wget git build-essential -y
4 sudo apt install virtualbox-guest-utils virtualbox-guest-x11
5 Y
6 y
7 Y
8 whoami
9 pwd
10 cat ?etc?os-ralease
11 cat /etc/os-release
12 mkdir -p ~/td_linux/{documents,sauvegardes,resultats}
13 ls -la ~/td_linux
14 cd ~/td_linux/documents
15 printf '%s\n' \ "Travaux sur linux" \ "mes commandes" \ "ubuntu" >notes.t
xt
16 ls
17 cd ..
18 cp documents/notes.txt sauvegardes/notes_initiales.txt
19 diff documents/notes.txt sauvegardes/notes_initiales.txt
20 mkdir temporaire

```
doris@Do:~/td_linux/documents$ cowsay "Welcome to linux"
 ____________________
< Welcome to linux >
 --------------------
        \   ^__^
         \  (oo)\_______
            (__)\       )\/\
                ||----w |
                ||     ||
doris@Do:~/td_linux/documents$
```




doris@Do: ~/td_linux/documents — journalctl
~/td_linux/documents
Oct 29 16:29:35 Do kernel: Linux version 7.0.0-42-generic (buildd@lcy02-amd64-0>
Oct 29 16:29:35 Do kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-7.0.0-42-gene>
Oct 29 16:29:35 Do kernel: KERNEL supported cpus:
Oct 29 16:29:35 Do kernel: Intel GenuineIntel
Oct 29 16:29:35 Do kernel: AMD AuthenticAMD
Oct 29 16:29:35 Do kernel: Hygon HygonGenuine
Oct 29 16:29:35 Do kernel: Centaur CentaurHauls
Oct 29 16:29:35 Do kernel: zhaoxin Shanghai
Oct 29 16:29:35 Do kernel: x86/CPU: Model not found in latest microcode list
Oct 29 16:29:35 Do kernel: BIOS-provided physical RAM map:
Oct 29 16:29:35 Do kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fff>
Oct 29 16:29:35 Do kernel: BIOS-e820: [gap 0x00000000000a0000-0x00000000000ffff>
Oct 29 16:29:35 Do kernel: BIOS-e820: [mem 0x0000000000100000-0x00000000ddea2ff>
Oct 29 16:29:35 Do kernel: BIOS-e820: [mem 0x00000000ddea3000-0x00000000ddfb6ff>
Oct 29 16:29:35 Do kernel: BIOS-e820: [mem 0x00000000ddfb7000-0x00000000deaecff>
Oct 29 16:29:35 Do kernel: BIOS-e820: [mem 0x00000000deaed000-0x00000000ded6cff>
Oct 29 16:29:35 Do kernel: BIOS-e820: [mem 0x00000000ded6d000-0x00000000ded7eff>
Oct 29 16:29:35 Do kernel: BIOS-e820: [mem 0x00000000ded7f000-0x00000000dedfeff>
Oct 29 16:29:35 Do kernel: BIOS-e820: [mem 0x00000000dedff000-0x00000000df12cff>
Oct 29 16:29:35 Do kernel: BIOS-e820: [mem 0x00000000df12d000-0x00000000df130ff>
Oct 29 16:29:35 Do kernel: BIOS-e820: [mem 0x00000000df131000-0x00000000df132ff>
Oct 29 16:29:35 Do kernel: BIOS-e820: [mem 0x00000000df133000-0x00000000df16dff>
Oct 29 16:29:35 Do kernel: BIOS-e820: [mem 0x00000000df16e000-0x00000000dffefff>
lines 1-23


doris@Do: ~/td_linux/documents
~/td_linux/documents
efff]
[0.080236] PM: hibernation: Registered nosave memory: [mem 0xdf12d000-0xdf13
2fff]
[0.080237] PM: hibernation: Registered nosave memory: [mem 0xdf16e000-0xffff
ffff]
[0.080267] Early memory node ranges
[0.166409] Freeing SMP alternatives memory: 56K
[0.280360] Memory: 2470148K/5214084K available (23912K kernel code, 4961K rw
data, 16720K rodata, 5316K init, 5824K bss, 2736012K reserved, 0K cma-reserved)
[0.281224] x86/mm: Memory block size: 128MB
[0.285714] efi: Freeing EFI boot services memory: 25192K
[0.653604] Freeing initrd memory: 37732K
[1.162091] memory_tiering memory_tier4: hash matches
[1.164449] Freeing unused decrypted memory: 2028K
[1.165220] Freeing unused kernel image (initmem) memory: 5316K
[1.165828] Freeing unused kernel image (text/rodata gap) memory: 660K
[1.166398] Freeing unused kernel image (rodata/data gap) memory: 1712K
[1.688196] vmwgfx 0000:00:02.0: [drm] Legacy memory limits: VRAM = 16384 KiB
, FIFO = 2048 KiB, surface = 507904 KiB
[1.688208] vmwgfx 0000:00:02.0: [drm] Maximum display memory size is 16384 K
iB
[4.946342] systemd[1]: Listening on systemd-oomd.socket - Userspace Out-Of-M
emory (OOM) Killer Socket.
doris@Do:~/td_linux/documents$


doris@Do: ~/td_linux/documents
~/td_linux/documents
Summary:
Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 3
Download size: 20.8 kB
Space needed: 59.4 kB / 11.7 GB available
Get:1 http://ht.archive.ubuntu.com/ubuntu resolute/universe amd64 ncal amd64 12.
1.8build1 [20.8 kB]
Fetched 20.8 kB in 1s (26.7 kB/s)
Selecting previously unselected package ncal.
(Reading database… 236226 files and directories currently installed.)
Preparing to unpack …/ncal_12.1.8build1_amd64.deb…
Unpacking ncal (12.1.8build1)…
Setting up ncal (12.1.8build1)…
Processing triggers for man-db (2.13.1-1build1)…
doris@Do:~/td_linux/documents$ cal
November 2026
Su Mo Tu We Th Fr Sa
1 2 3 4 5 6 7
8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30
doris@Do:~/td_linux/documents$


doris@Do:~/td_linux/documents$ figlet "UBUNTU"

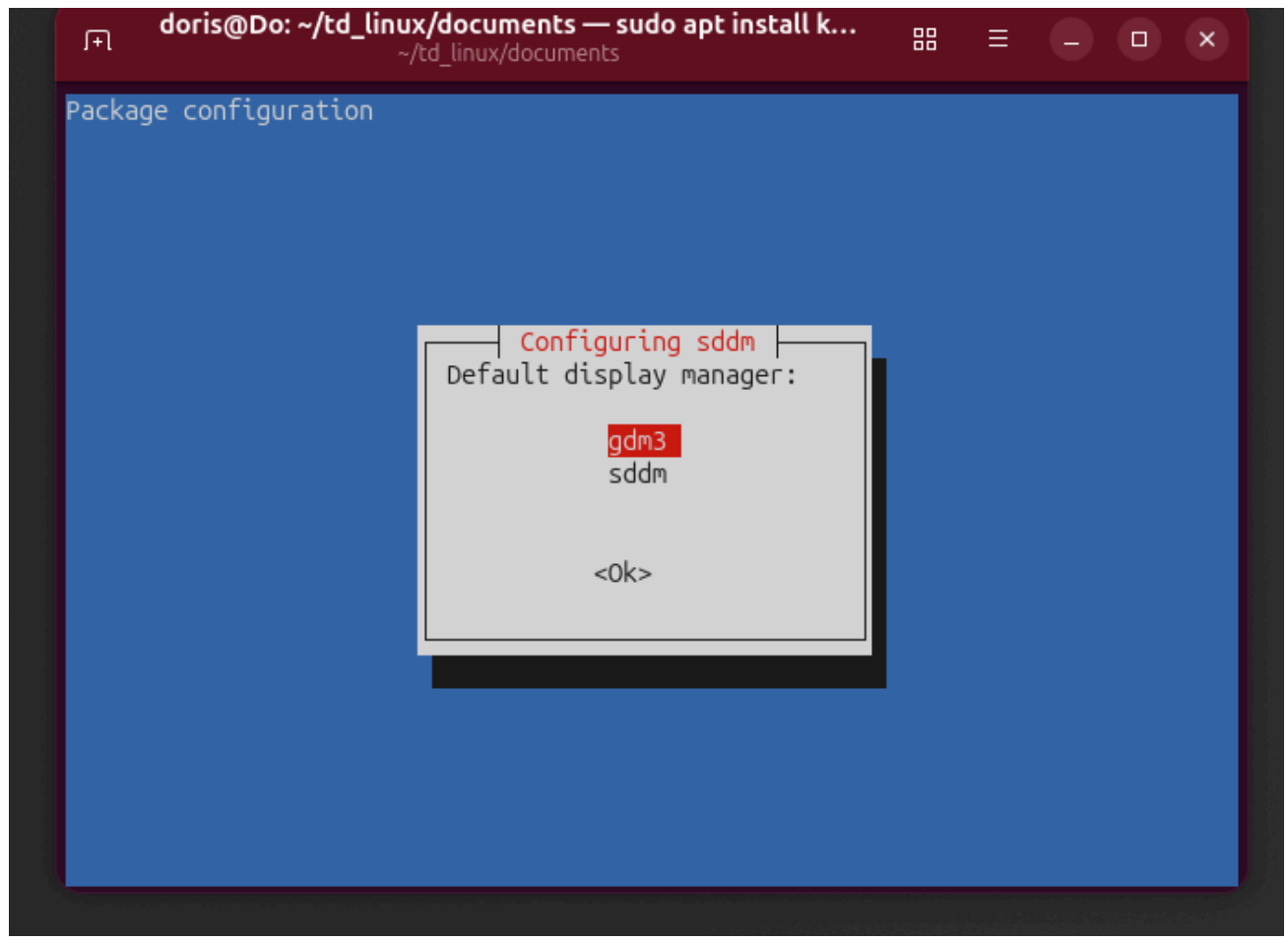

~/td_linux/documents
etting up sl (5.02-1build1)…
rocessing triggers for man-db (2.13.1-1build1)…
oris@Do:~/td_linux/documents$ seq 1 10
0
oris@Do:~/td_linux/documents$ seq 2 2 20
0
2
4
6
8
0
6
7
8
9
10
doris@Do:~/td_linux/documents$ seq 2 2 20
2
4
6
8
10
12
14
16
18
20
doris@Do:~/td_linux/documents$ shuf -i 1-100 -n 1
79
doris@Do:~/td_linux/documents$

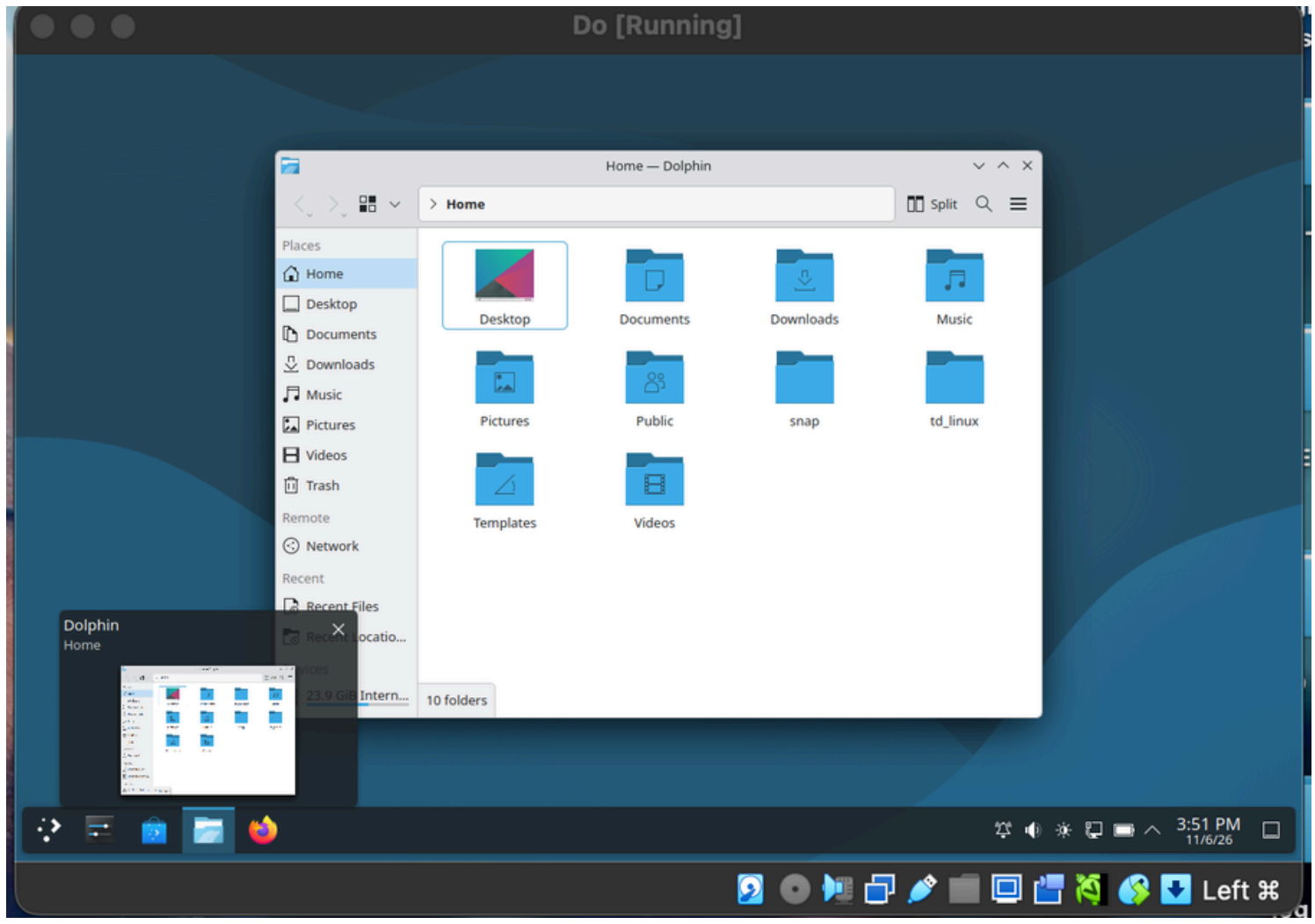
```
doris@Do:~/td_linux/documents$ shuf -i 1-100 -n 1
79
doris@Do:~/td_linux/documents$ echo "$XDG_CURRENT_DESKTOP"
ubuntu:GNOME
doris@Do:~/td_linux/documents$ echo "$XDG_SESSION_TYPE"
wayland
doris@Do:~/td_linux/documents$ apt policy kde-plasma-desktop xfce4
kde-plasma-desktop:
  Installed: (none)
  Candidate: 5:166ubuntu2
  Version table:
     5:166ubuntu2 500
        500 http://ht.archive.ubuntu.com/ubuntu resolute/universe amd64 Packages
xfce4:
  Installed: (none)
  Candidate: 4.20.1
  Version table:
     4.20.1 500
        500 http://ht.archive.ubuntu.com/ubuntu resolute/universe amd64 Packages
doris@Do:~/td_linux/documents$ apt -s install kde-plasma-desktop
NOTE: This is only a simulation!
      apt needs root privileges for real execution
```

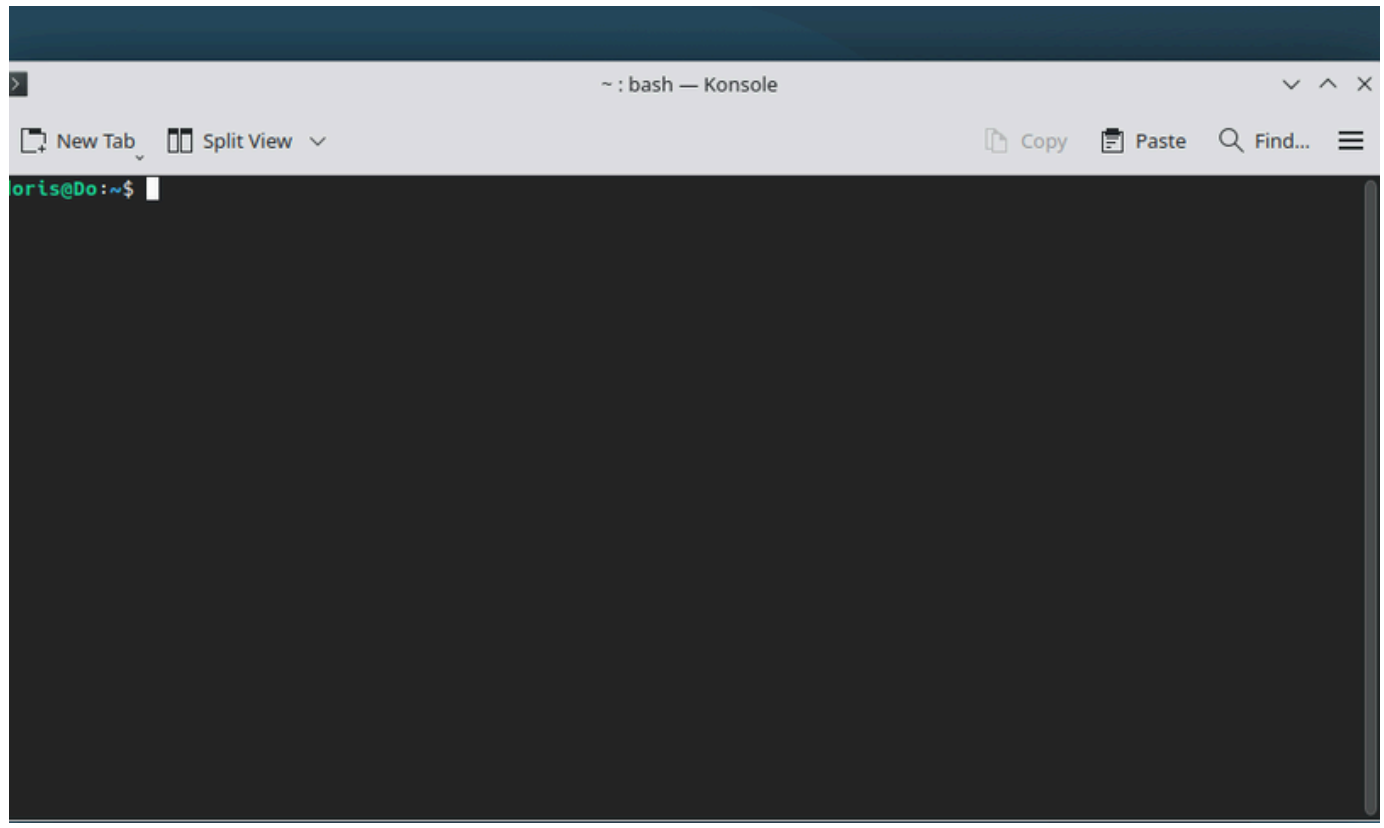
```
doris@Do: ~/td_linux/documents — sudo apt install k...
~/td_linux/documents
Conf xserver-xorg-legacy (2:21.1.22-1ubuntu1.3 Ubuntu:26.04/resolute-updates [am
d64])
doris@Do:~/td_linux/documents$ sudo apt install kde-plasma-desktop
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-30                  linux-modules-7.0.0-30-generic
  linux-headers-7.0.0-30-generic          linux-tools-7.0.0-30
  linux-image-unsigned-7.0.0-30-generic   linux-tools-7.0.0-30-generic
  linux-main-modules-zfs-7.0.0-30-generic
Use 'sudo apt autoremove' to remove them.

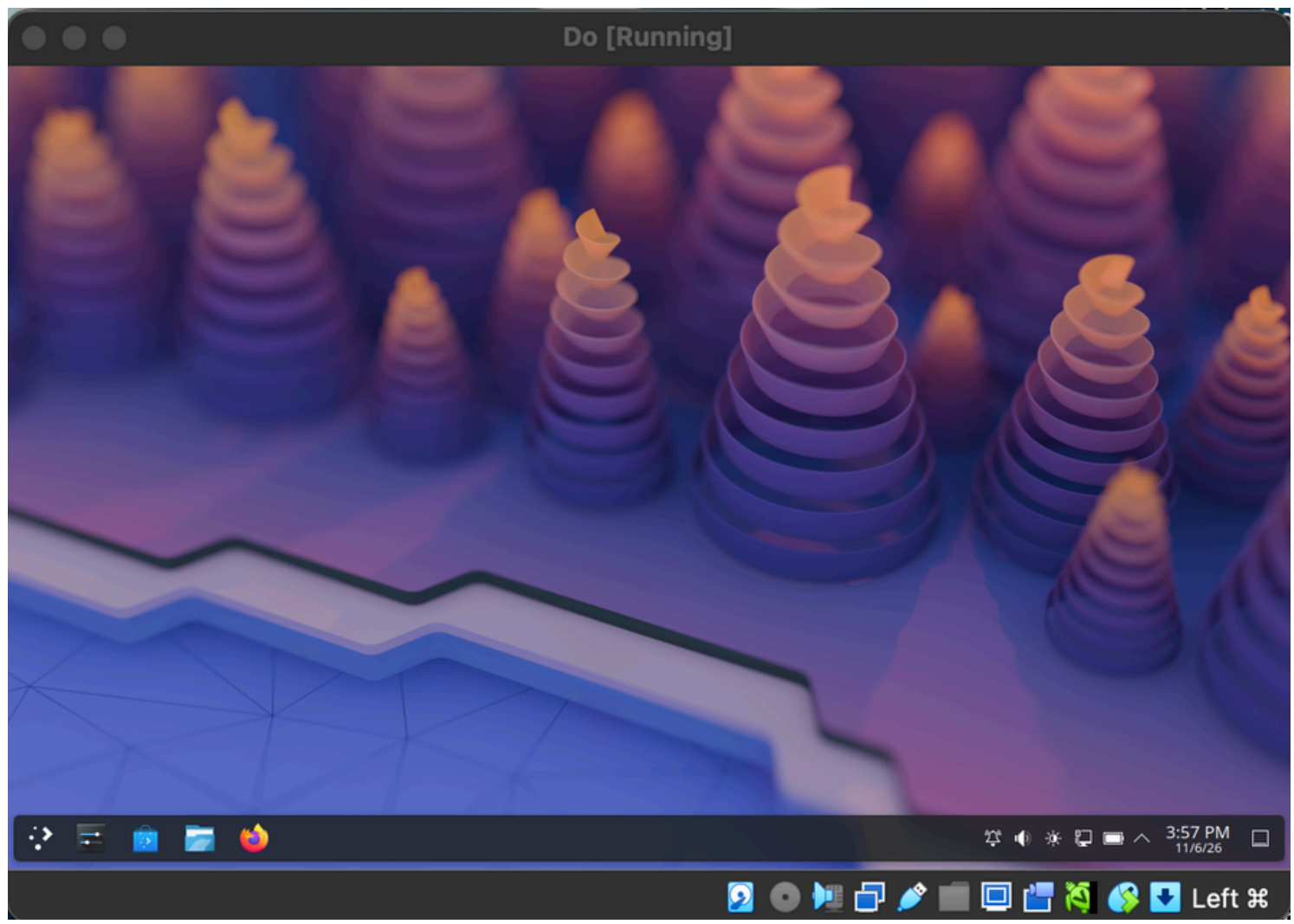
Installing:
  kde-plasma-desktop

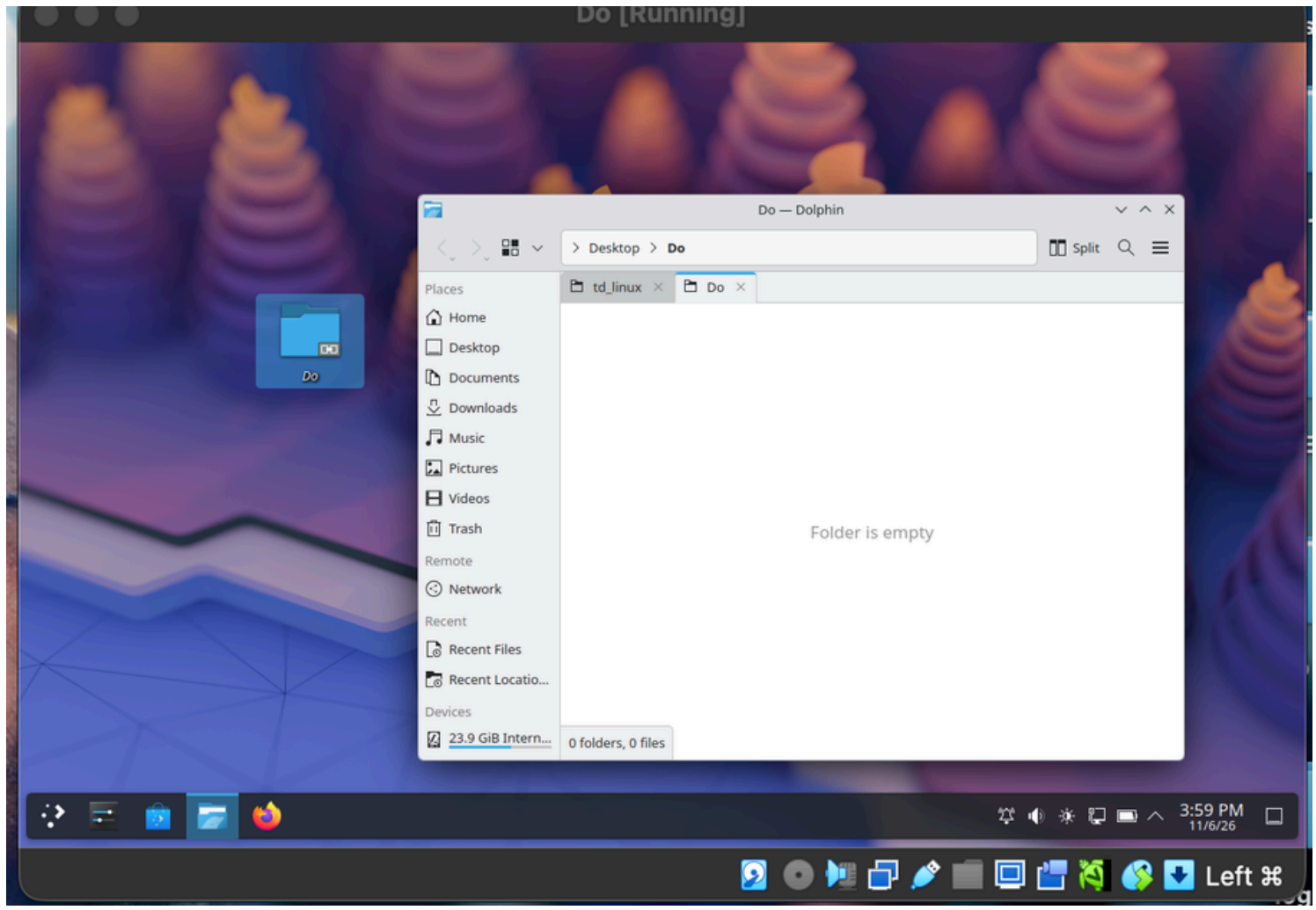
Installing dependencies:
  aha
  appmenu-gtk-module-common
  appmenu-gtk3-module
  apt-config-icons-large
  apt-config-icons-large-hidpi
  baloo6
  bluedevil
  breeze
  breeze-cursor-theme
  breeze-gtk-theme
```

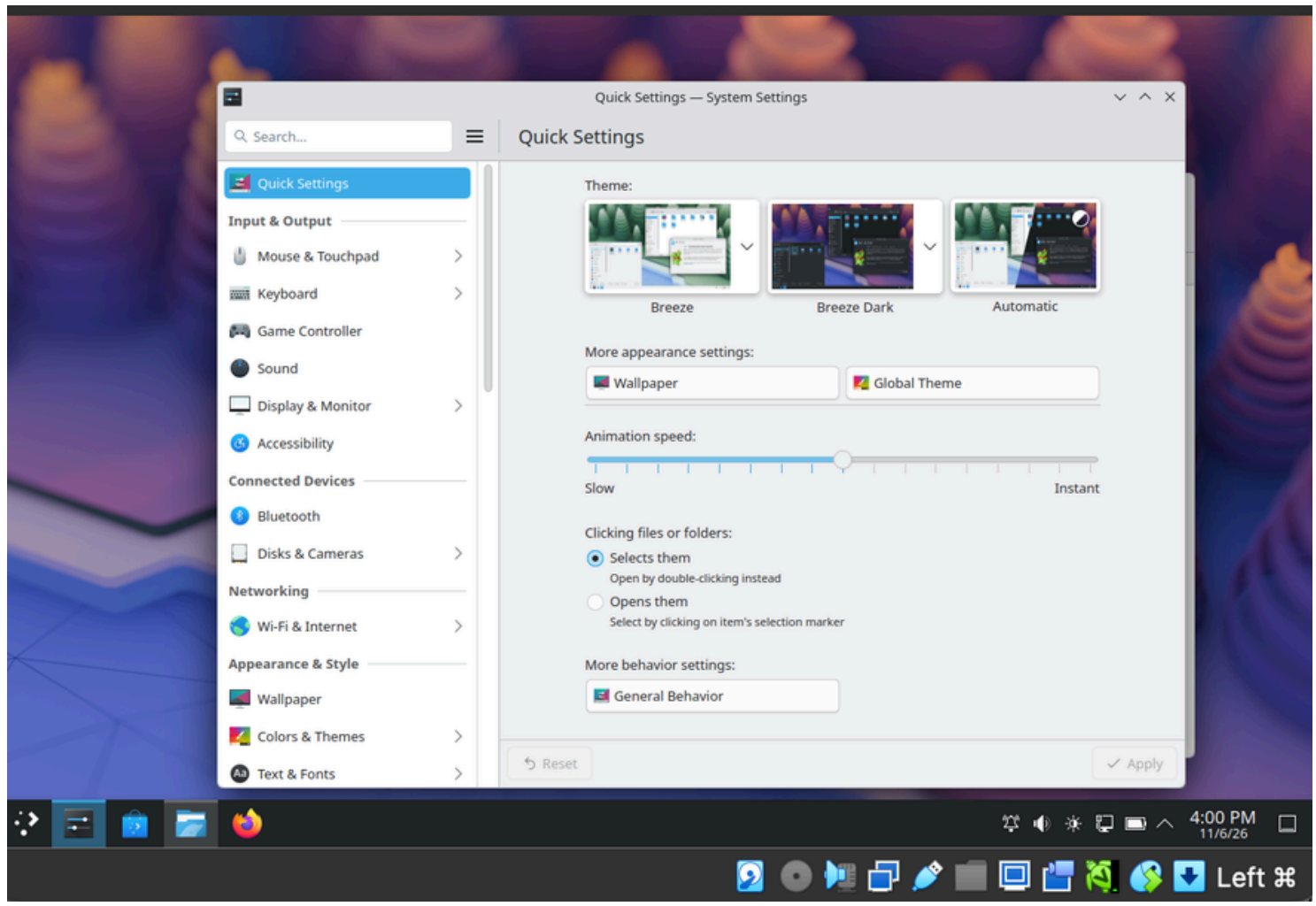

doris@Do: ~/td_linux/documents — sudo apt install k...
~/td_linux/documents
Package configuration
Configuring sddm
Default display manager:
gdm3
sddm
<Ok>


Do [Running]
Home — Dolphin
Home
Split
Places
Home
Desktop
Documents
Downloads
Music
Pictures
Videos
Trash
Remote
Network
Recent
Recent Files
Desktop
Documents
Downloads
Music
Pictures
Public
snap
td_linux
Templates
Videos
10 folders
Dolphin
Home
3:51 PM
11/6/26
Left ⌘


~ : bash — Konsole
New Tab
Split View
Copy
Paste
Find...
oris@Do:~$


Do [Running]
3:57 PM
11/6/26
Left ⌘

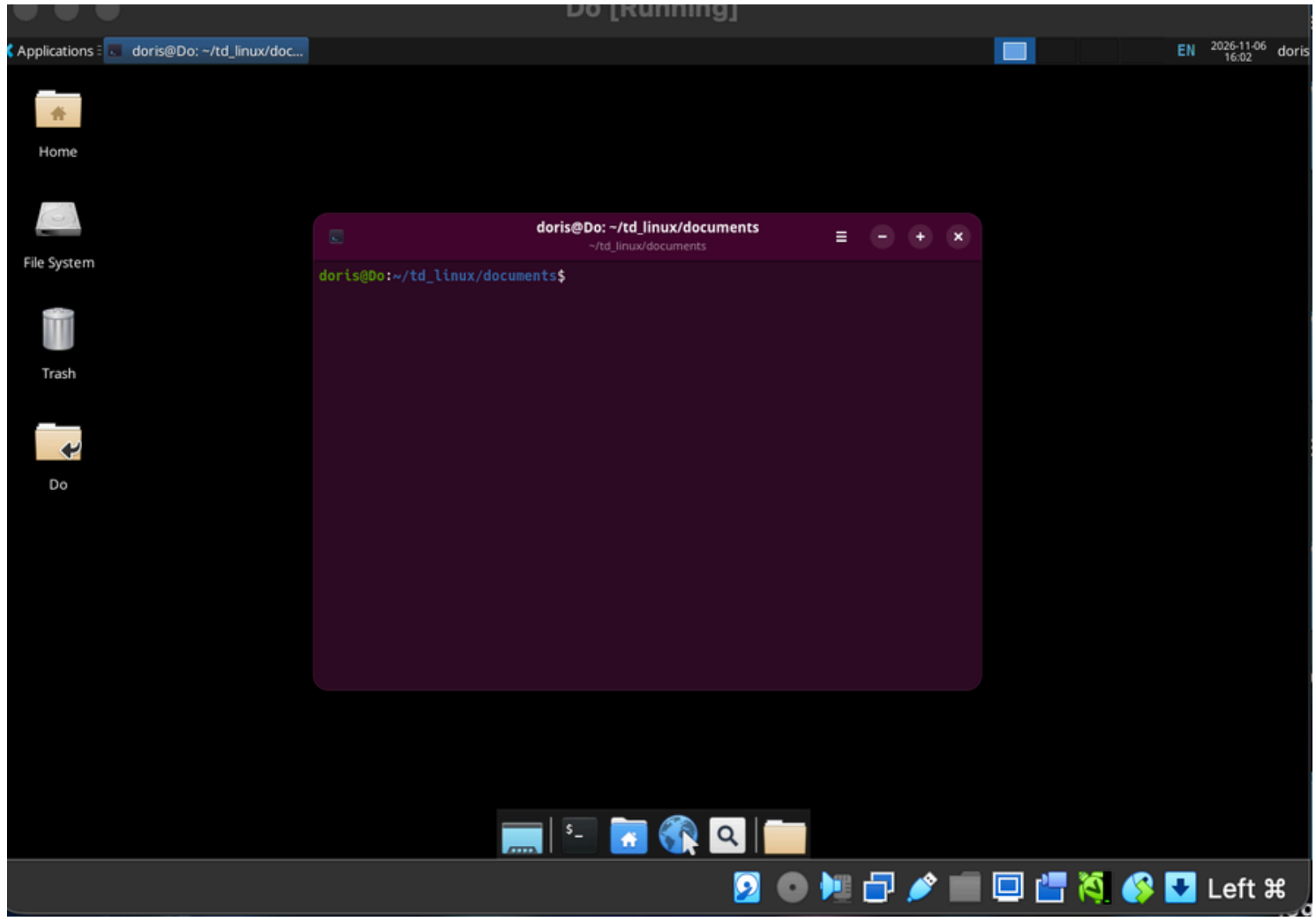

Do [Running]
Do — Dolphin
Desktop > Do
Split
Places
Home
Desktop
Documents
Downloads
Music
Pictures
Videos
Trash
Remote
Network
Recent
Recent Files
Recent Locatio...
Devices
23.9 GiB Intern...
td_linux
Do
Folder is empty
0 folders, 0 files
Do
3:59 PM
11/6/26
Left ⌘

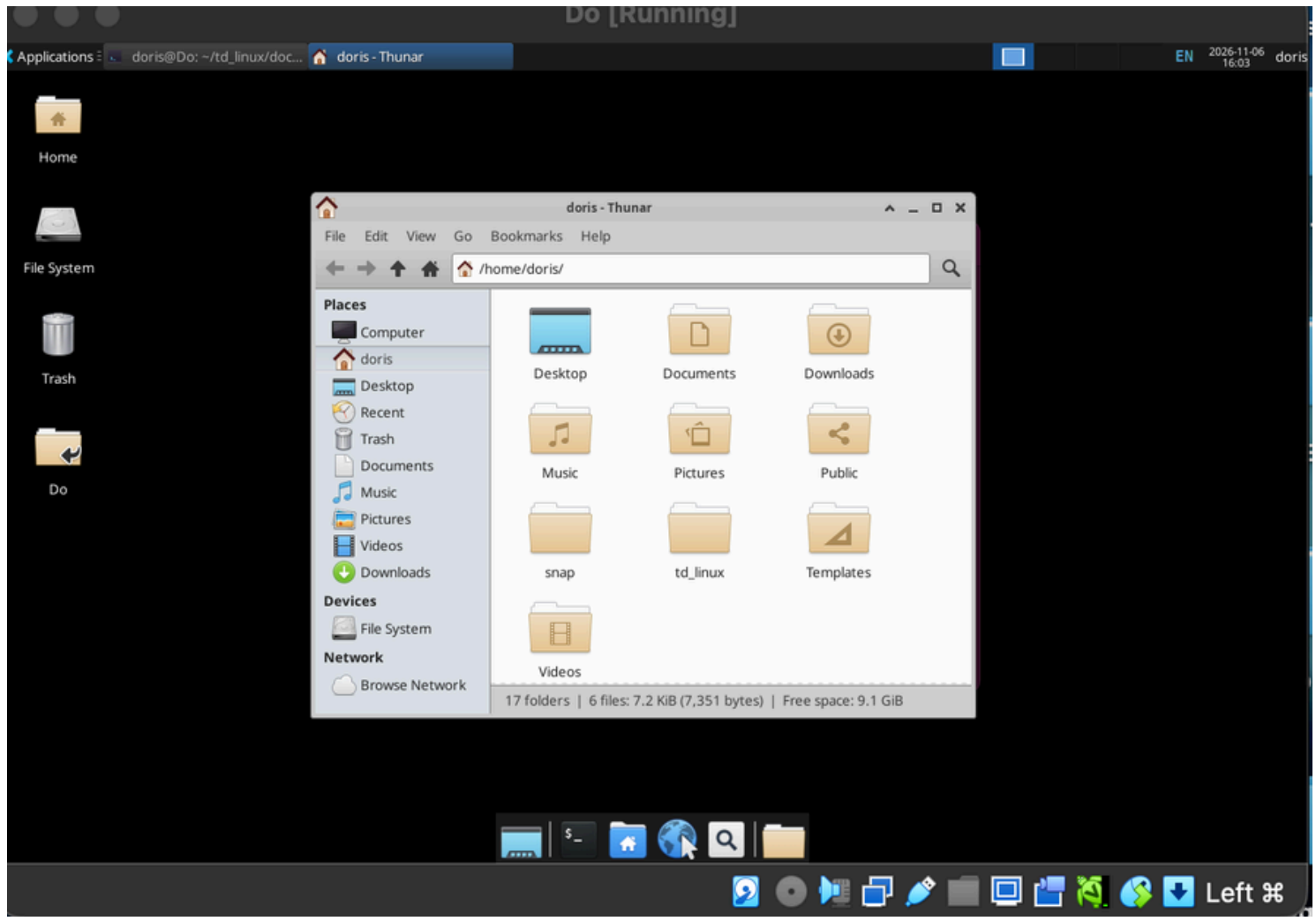

Quick Settings — System Settings
Search...
Quick Settings
Quick Settings
Input & Output
Mouse & Touchpad
Keyboard
Game Controller
Sound
Display & Monitor
Accessibility
Connected Devices
Bluetooth
Disks & Cameras
Networking
Wi-Fi & Internet
Appearance & Style
Wallpaper
Colors & Themes
Text & Fonts
Theme:
Breeze
Breeze Dark
Automatic
More appearance settings:
Wallpaper
Global Theme
Animation speed:
Slow
Instant
Clicking files or folders:
Selects them
Open by double-clicking instead
Opens them
Select by clicking on item's selection marker
More behavior settings:
General Behavior
Reset
Apply
4:00 PM
11/6/26
Left ⌘

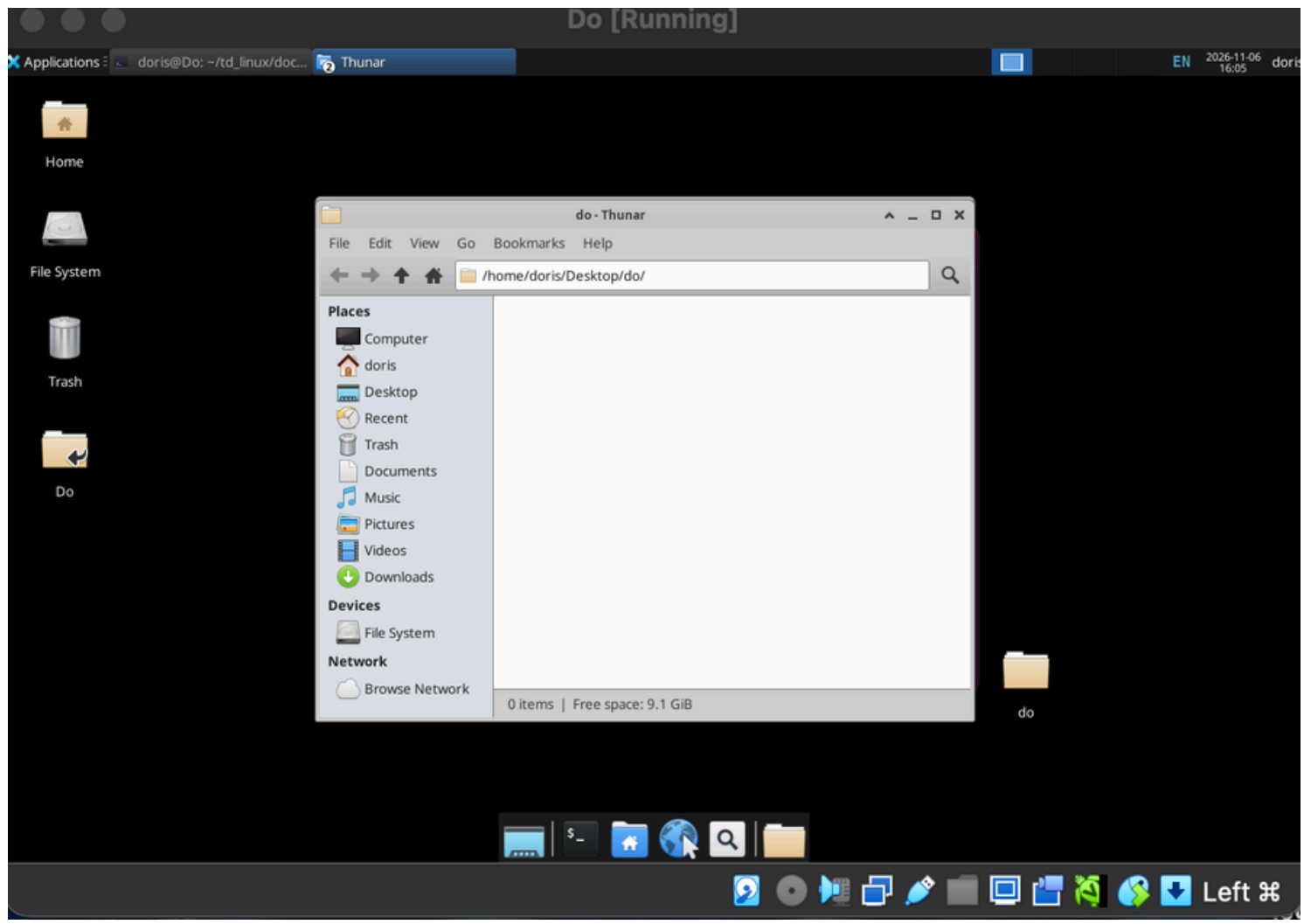
~ : bash — Konsole

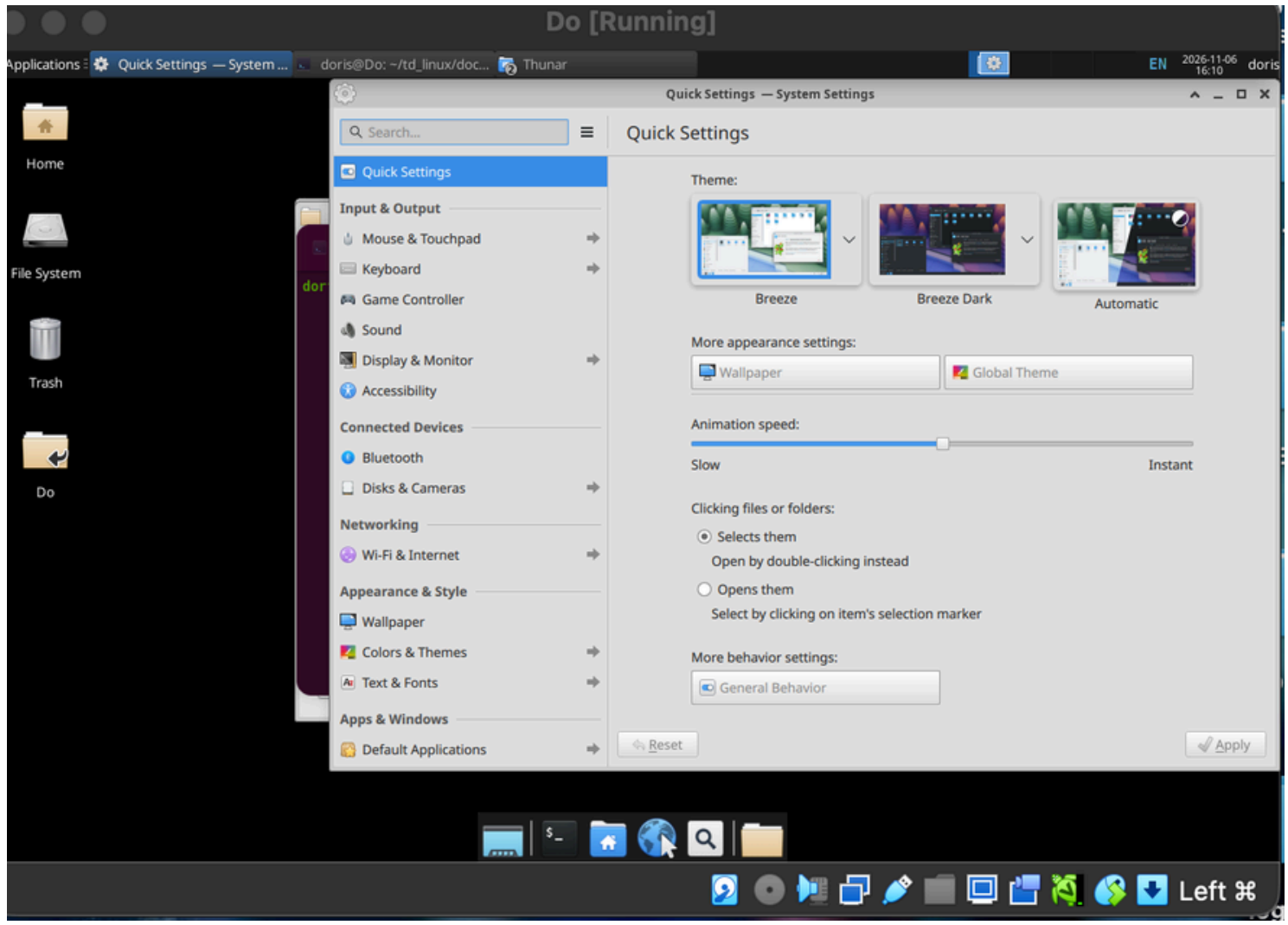
New Tab Split View Copy Paste Find...

```
doris@Do:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           4.3Gi       1.8Gi       229Mi        55Mi       2.6Gi       2.5Gi
Swap:          4.0Gi       8.0Ki       4.0Gi
doris@Do:~$
```

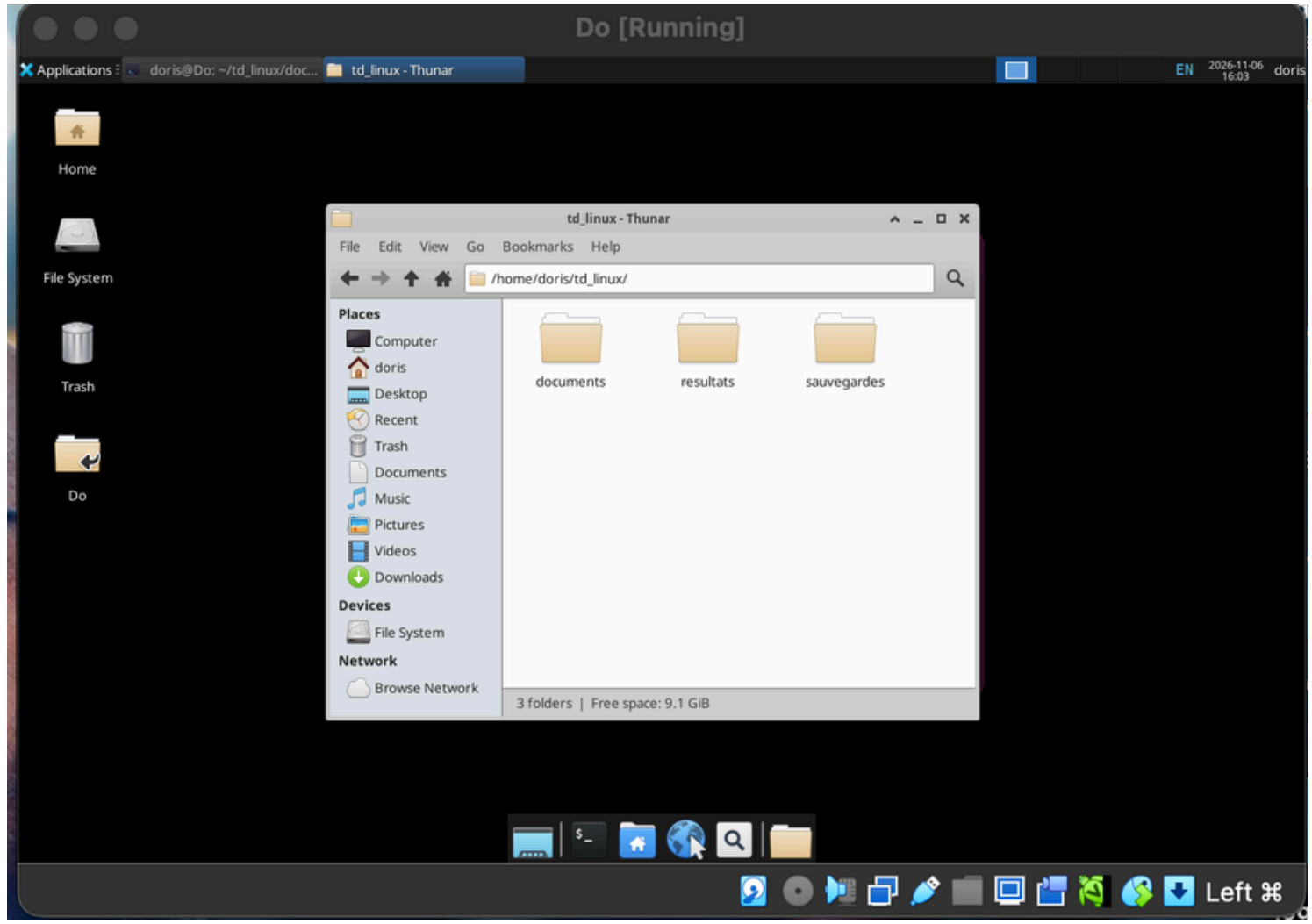



Do [Running]
Applications
doris@Do: ~/td_linux/doc...
doris - Thunar
EN
2026-11-06
16:03
doris
Home
File System
Trash
Do
doris - Thunar
File Edit View Go Bookmarks Help
/home/doris/
Places
Computer
doris
Desktop
Recent
Trash
Documents
Music
Pictures
Videos
Downloads
Devices
File System
Network
Browse Network
Desktop
Documents
Downloads
Music
Pictures
Public
snap
td_linux
Templates
Videos
17 folders | 6 files: 7.2 KiB (7,351 bytes) | Free space: 9.1 GiB
Left ⌘


Do [Running]
Applications
doris@Do: ~/td_linux/doc...
Thunar
EN
2026-11-06
16:05
doris
Home
File System
Trash
Do
do - Thunar
File Edit View Go Bookmarks Help
/home/doris/Desktop/do/
Places
Computer
doris
Desktop
Recent
Trash
Documents
Music
Pictures
Videos
Downloads
Devices
File System
Network
Browse Network
0 items | Free space: 9.1 GiB
do
Left ⌘


Do [Running]
Applications
Quick Settings — System ...
doris@Do: ~/td_linux/doc...
Thunar
EN
2026-11-06
16:10
doris
Home
File System
Trash
Do
Quick Settings — System Settings
Search...
Quick Settings
Quick Settings
Input & Output
Mouse & Touchpad
Keyboard
Game Controller
Sound
Display & Monitor
Accessibility
Connected Devices
Bluetooth
Disks & Cameras
Networking
Wi-Fi & Internet
Appearance & Style
Wallpaper
Colors & Themes
Text & Fonts
Apps & Windows
Default Applications
Theme:
Breeze
Breeze Dark
Automatic
More appearance settings:
Wallpaper
Global Theme
Animation speed:
Slow
Instant
Clicking files or folders:
Selects them
Open by double-clicking instead
Opens them
Select by clicking on item's selection marker
More behavior settings:
General Behavior
Reset
Apply
Left ⌘

Applications
Quick Settings * — System...
doris@Do: ~/td_linux/doc...
Thunar
EN
2026-11-06
16:11
doris
Home
File System
Trash
Do
Quick Settings * — System Settings
Search...
Quick Settings
Quick Settings
Input & Output
Theme:
Dark
Automatic
Global Theme
Instant
Apps & Windows
Default Applications
Reset
Apply
doris@Do: ~/td_linux/documents
~/td_linux/documents
doris@Do:~/td_linux/documents$ free -h
total used free shared buff/cache available
Mem: 4.3Gi 1.2Gi 462Mi 20Mi 2.9Gi 3.1Gi
Swap: 4.0Gi 8.0Ki 4.0Gi
doris@Do:~/td_linux/documents$
Left ⌘


Do [Running]
Applications
doris@Do: ~/td_linux/doc...
td_linux - Thunar
EN
2026-11-06
16:03
doris
Home
File System
Trash
Do
td_linux - Thunar
File Edit View Go Bookmarks Help
/home/doris/td_linux/
Places
Computer
doris
Desktop
Recent
Trash
Documents
Music
Pictures
Videos
Downloads
Devices
File System
Network
Browse Network
documents
resultats
sauvegardes
3 folders | Free space: 9.1 GiB
Left ⌘

Critère	KDE Plasma	XFCE
Terminal utilisé	Konsole	XFCE Terminal
Gestionnaire de fichiers	Dolphin	Thunar
Acces aux paramètres	Via Configuration du système (System Settings).	Via Gestionnaire de paramètres (Settings Manager).
Personnalisation	Très avancée : thèmes, icônes, widgets, effets visuels et panneaux.	Simple et flexible : thèmes, icônes, panneaux et fond d'écran.
Mémoire disponible	À mesurer avec free -h.	À mesurer avec free -h.
Choix pour un poste donné	Adapté aux ordinateurs disposant de bonnes ressources et recherchant une interface moderne et complète.	Adapté aux ordinateurs ayant des ressources limitées et recherchant un environnement léger et rapide.